Technical Report: UR3e Cobot Early Activity (April 14)

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Course: ITAI 2374 – Spring 2026

Project Activities

Our team's project focused on the foundational deployment of the **UR3e Cobot** in **Robotics Lab N138**. Following the **UR20/30 E-Learning** path guided by the virtual supervisor, Bryan, we completed the first four curriculum modules:

- **Hardware Overview and Setting Up (Module 1):**
  We performed a full system initialization, including powering up the robot, navigating the **Teach Pendant**, and exploring the **Control Box**. We also successfully mounted and connected the end-effector (gripper) and practiced using the **3PE switch** to move the robot.

- **Configuring a Tool (Module 2):**
  We focused on essential configuration procedures for the mounted gripper. This included teaching the **Tool Center Point (TCP)**, setting up the **Orientation configuration**, and defining the **Payload and Center of Gravity** to ensure accurate robotic movement.

- **Movement and Motion (Module 3):**
  Our team learned about the robot's distinct motion types. We practiced moving the robot directly from the **Move screen** and learned to use the **Align tool** and **Home position** features.

- **Creating a Program (Module 4):**
  We transitioned into basic programming by Setting Waypoints, which is considered the most critical skill in the tutorial. We also practiced **renaming waypoints**, setting up a **Before start-sequence**, and managing **Program folders**.

Challenges and Obstacles

- **TCP Calibration:** During the tool configuration in Module 2, we initially struggled with the **Orientation configuration**, which is necessary for the robot to understand its tool's position in 3D space.
    - **Solution:** We revisited the **"Configuration of a Tool Using the Wizard"** video tutorial to ensure our manual TCP inputs were precise.

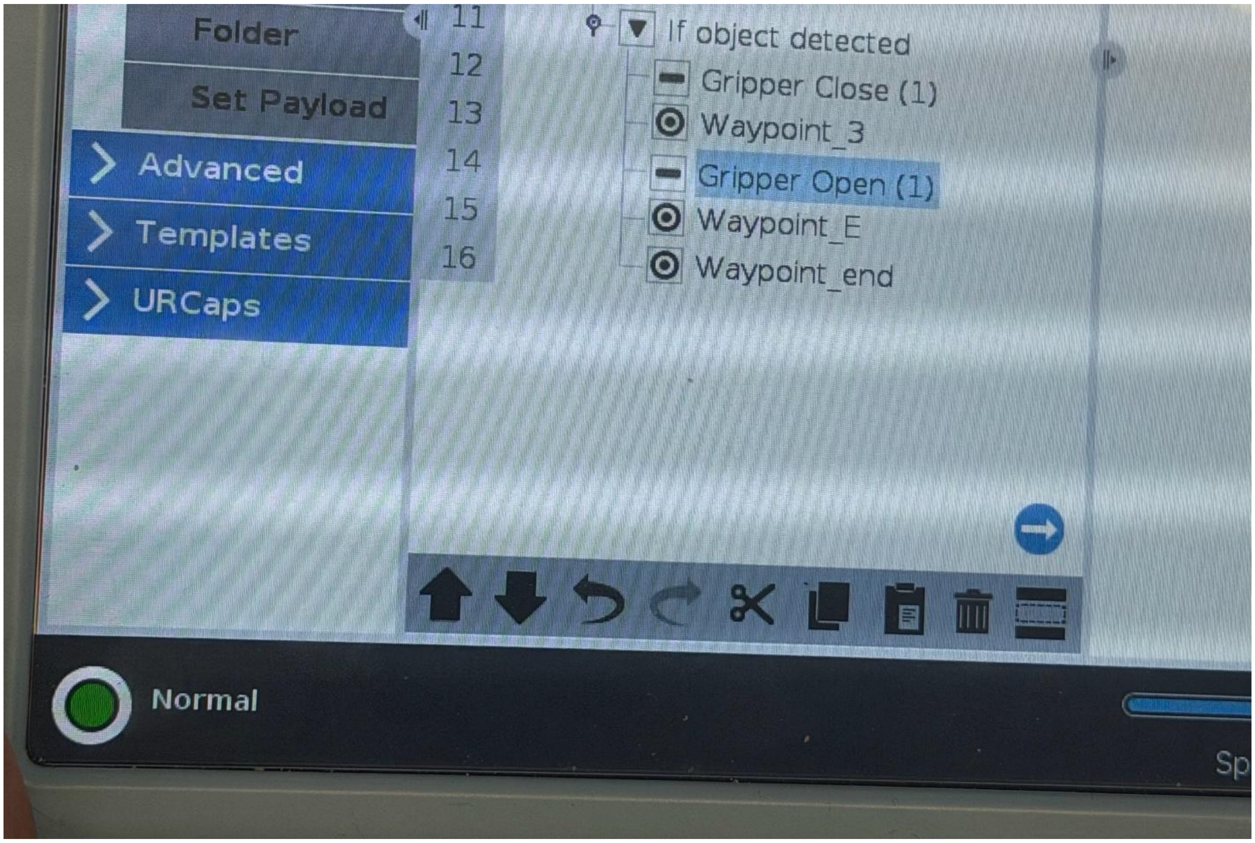
- **Access Controls:** Navigating the software required specific permissions that were not immediately obvious.
    - **Solution:** We learned to use the **Operational mode password** as outlined in the Module 4 training to unlock necessary programming features.

Accomplishments

Our team successfully transitioned from the E-learning simulations to a functional physical setup in Lab N138. We demonstrated a precise vertical approach and retreat sequence using the following setup:

- **Physical Configuration:** We integrated the UR3e with a Hand-E Gripper and aligned it with the Dorner 2200 Series conveyor.
- **Programmed Sequence:** Using the skills from Module 4, we created a program that starts at a "Home" position, moves to a "Hover" waypoint, and executes a controlled descent.
- **Result:** The robot kept perfect perpendicular alignment throughout the motion, confirming our TCP and orientation settings were correct.

Figure 1: Our physical lab setup showing the UR3e arm equipped with the Hand-E gripper, positioned for interaction with the Dorner conveyor belt.



Video 1: A 10-second demonstration of the programmed waypoint sequence and smooth vertical motion.

https://youtube.com/shorts/oE1sZEHLwzY?si=-hUoOL8lHpCqD9ed

Learning Outcomes

From these activities, our team learned the vital connection between software configuration and physical safety. We realized that if the Payload and Center of Gravity are not set correctly in the early stages, the robot's motion becomes jerky or triggers safety stops. We also learned that using Free Drive is the most efficient way to establish a "rough" path before fine-tuning the coordinates on the Teach Pendant for the final pick-and-place points.

Critical Analysis

We found the UR3e equipment and the Universal Robots Academy video instructions to be very appropriate for this course level. The videos provided a safe, simulated environment to learn about the GUI before we touched the actual hardware. While the videos used a newer UR20 model, the **PolyScope GUI** logic was nearly identical, making it a highly effective instructional tool for learning how to establish waypoints.